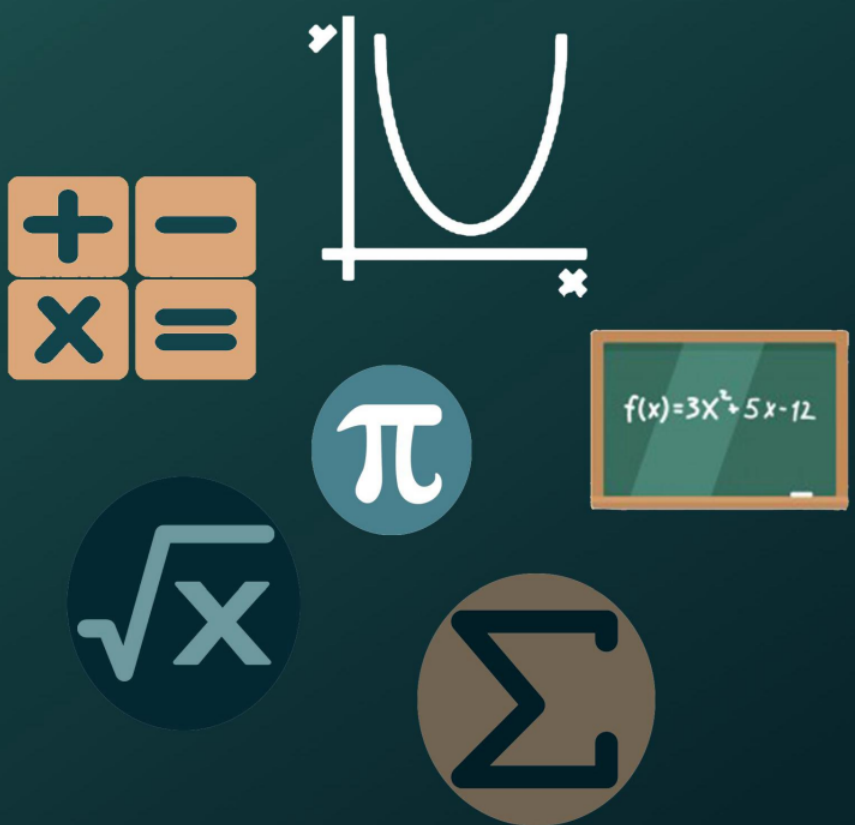
MATHEMATICS

# Definite Integration




## SHEET

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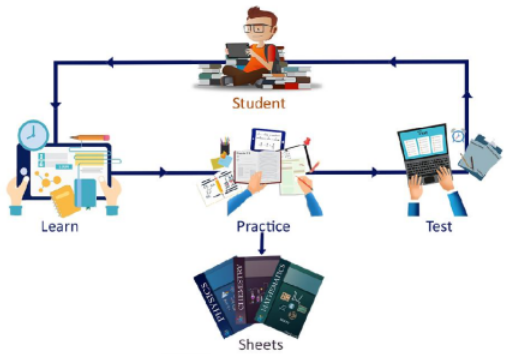
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***Note***

*Detailed solutions are available on the eSaral App.*

# *Module Description*




For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

*"Practice makes a man perfect"*

**1. Concept builder – 1 & 1A**

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

**2. Brain Booster – 2 & 2A**

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

**3. Simulator – JM & JA**

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

# *Home Work Index*

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

| S.No | Topic Name | Ex 1 | Ex 1A | Ex 2 | Ex 2A |
|---|---|---|---|---|---|
| 1 | Geometrical Interpretation of Definite Integration | 1-8 | 13-15, 19 | | |
| 2 | Evaluating Definite Integrals by Finding Antiderivatives | | 10-12,23,34-39 | 1-6 | 18, 21, 22, 26-32 |
| 3 | Invalid Substitution | | 40-41 | | |
| 4 | Properties of Definite Integration-Property 1, 2 & 3 | 9-12 | 1-2,20,24-26 | 7,8,18,21 | 1, 2, 8-11, 19, 23-25,33-38 |
| 5 | Walli's Theorem | 13 | 5-6,42-44 | 9-11 | |
| 6 | Properties of Definite Integration-Property 4 | 14-20 | 3-4,21,27 | 19,22 | 3, 39, 40 |
| 7 | Properties of Definite Integration-Property 5 | 21-24 | 22,45-48 | 12-15,23 | 41, 42 |
| 8 | Properties of Definite Integration-Property 6 | 25 | | 20 | 43 |
| 9 | Properties of Definite Integration-Property 7 | 26-27 | | | 4, 44 |
| 10 | Properties of Definite Integration-Property 8 | | 7 | | |
| 11 | Derivative of Antiderivatives (Newton – Leibnitz Formula) | 28 | 8-9,28-29,49 | 16, 24-26 | 5, 12-14, 45 |
| 12 | Sum of Series using Definite Integration | 29-30 | 30,50 | 17, 27-29 | 6,15-17, 20, 46 |
| 13 | Reduction Formula | 31-32 | 31, 32 | 30, 31 | 7 |
| 14 | After complete chapter | | 16-18, 33 | | 51 |

## Exercise - 1 — Concept Builder

### SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

### Elementary Integrals

**Q1.** $\int_0^{\pi/2} e^x \sin x \, dx =$

(A) $\frac{1}{2}(e^{\pi/2}-1)$ (B) $\frac{1}{2}(e^{\pi/2}+1)$

(C) $\frac{1}{2}(1-e^{\pi/2})$ (D) $2(e^{\pi/2}+1)$

**Q2.** $\int_0^{\pi/2} \sqrt{\cos\theta}\sin^3\theta \, d\theta$

(A) $\frac{20}{21}$ (B) $\frac{8}{21}$

(C) $\frac{-20}{21}$ (D) $\frac{-8}{21}$

**Q3.** $\int_0^{\pi/2} \frac{\sin x \cos x}{1+\sin^4 x} dx =$

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$ (D) $\frac{\pi}{8}$

**Q4.** The solution for x of the equation $\int_{\sqrt{2}}^{x} \frac{dt}{t\sqrt{t^2-1}} = \frac{\pi}{12}$ is-

(A) 2 (B) $\pi$

(C) $\sqrt{3}/2$ (D) $2\sqrt{2}$

**Q5.** The value of $\int_1^2 \frac{dx}{x(1+x^4)}$ is-

(A) $\frac{1}{4}\log\frac{17}{32}$ (B) $\frac{1}{4}\log\frac{17}{2}$

(C) $\log\frac{17}{2}$ (D) $\frac{1}{4}\log\frac{32}{17}$

**Q6.** If $g(1) = g(2)$ then the value of $\int_1^2 [f\{g(x)\}]^{-1} f'\{g(x)\}g'(x)\,dx$ is-

(A) 1 (B) 2

(C) 0 (D) none of these

**Q7.** $\int_1^{e^{37}} \frac{\pi\sin(\pi\ell nx)}{x} dx =$

(A) 1 (B) 2

(C) e (D) 37

**Q8.** $\int_0^{\pi/2} \{x-[\sin x]\}dx$ is equal to (where [.] denotes the greatest integer)-

(A) $\frac{\pi^2}{8}$ (B) $\frac{\pi^2}{8}-1$

(C) $\frac{\pi^2}{8}-2$ (D) none of these

### Properties of Definite Integration
### Property 1, 2 & 3

**Q9.** If $\int_0^{\pi} |\cos x| dx =$

(A) $\pi$ (B) 0

(C) 2 (D) 1

**Q10.** $\int_{-4}^{4} |x+2| dx =$

(A) 50 (B) 24

(C) 20 (D) none of these

**Q11.** $\int_0^{\sqrt{2}} [x^2]\,dx$ is equal to (where [.] denotes greatest integer function)

(A) $\sqrt{2}-1$ (B) $2(\sqrt{2}-1)$

(C) $\sqrt{2}$ (D) none of these

**Q12.** The value of integral $\int_0^{\log 5} \frac{e^x\sqrt{e^x-1}}{e^x+3}dx =$

(A) $3+2\pi$ (B) $4-\pi$

(C) $2+\pi$ (D) none of these

## Walli's Theorem

**Q13.** $\int_0^{\pi/2} \sin^5 x\,dx =$

(A) $\frac{8}{15}$ (B) $\frac{4}{15}$

(C) $\frac{8\sqrt{\pi}}{15}$ (D) $\frac{8\pi}{15}$

## Properties of Definite Integration -Property 4

**Q14.** $\int_{-1}^{1} \frac{x^3\sin(1+x^2)}{1+x^2}dx$ is equal to-

(A) 1 (B) 2

(C) 0 (D) none of these

**Q15.** $\int_{-\pi/2}^{\pi/2} \cos^2 x\,dx$ is equal to

(A) $\pi/4$ (B) $\pi/2$

(C) $\pi/6$ (D) $\pi/3$

**Q16.** The value of

$\int_{-1}^{1} (\sqrt{1+x+x^2}-\sqrt{1-x+x^2})\,dx$ is

(A) 0 (B) 1

(C) $-1$ (D) None of these

**Q17.** If $f(x)$ is an odd function of $x$, then $\int_{-\pi/2}^{\pi/2} f(\cos x)\,dx$ is equal to

(A) 0 (B) $\int_0^{\pi/2} f(\cos x)\,dx$

(C) $2\int_0^{\pi/2} f(\sin x)\,dx$ (D) $\int_0^{\pi} f(\cos x)\,dx$

**Q18.** $\int_{-\pi/3}^{\pi/3} \frac{x^3\cos x}{\sin^2 x}dx =$

(A) 0 (B) 1

(C) $-1$ (D) none of these

**Q19.** $\int_{-1}^{1} \sin^{-1}\frac{x}{1+x^2}dx =$

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$

(C) $\pi$ (D) 0

**Q20.** If $f: R \to R$ and $g: R \to R$ are one-one real function, then

$\int_{-\pi}^{\pi} [f(x)+f(-x)][g(x)-g(-x)]\,dx =$

(A) 0 (B) $\pi$

(C) 1 (D) none of these

## Properties of Definite Integration - Property 5

**Q21.** $\int_0^1 f(1-x)dx$ is equal to-

(A) $\int_0^1 f(x)dx$ (B) $\int_0^1 f(-x)dx$

(C) $\int_0^1 f(x-1)dx$ (D) $\int_{-1}^1 f(x)dx$

**Q22.** $\int_0^{\pi} x \sin x \, dx =$

(A) $\pi$ (B) 0
(C) 1 (D) $\pi^2$

**Q23.** If $f(a+b-x) = f(x)$, then $\int_a^b x f(x) \, dx$ is equal to-

(A) $\frac{a+b}{2}\int_a^b f(a-x) \, dx$

(B) $\frac{a+b}{2}\int_a^b f(b-x) \, dx$

(C) $\frac{a+b}{2}\int_a^b f(x) \, dx$

(D) $\frac{b-a}{2}\int_a^b f(x) \, dx$

**Q24.** $\int_0^{\pi} x f(\sin x) \, dx =$

(A) $\pi\int_0^{\pi} f(\cos x) \, dx$ (B) $\frac{\pi}{2}\int_0^{\pi} f(\sin x) \, dx$

(C) $\frac{\pi}{2}\int_0^{\pi/2} f(\cos x) \, dx$ (D) None of these

## Properties of Definite Integration 6

**Q25.** If $\int_0^{\pi} x f(\cos^2 x + \tan^4 x) \, dx$

$= k\int_0^{\pi/2} f(\cos^2 x + \tan^4 x) \, dx$

then, the values of k is-

(A) $\frac{\pi}{2}$ (B) $\pi$

(C) $-\frac{\pi}{2}$ (D) none of these

## Properties of Definite Integration 7

**Q26.** $\int_{\pi}^{10\pi} |\sin x| \, dx =$

(A) 9 (B) 10
(C) 18 (D) 20

**Q27.** The value of $\int_0^{100\pi} \sqrt{1-\cos 2x} \, dx$ is -

(A) $100\sqrt{2}$

(B) $200\sqrt{2}$

(C) $50\sqrt{2}$

(D) 0

## Derivatives of Antiderivatives (Newton - Leibnitz Formula)

**Q28.** The value of $\lim_{x\to 0} \frac{\int_0^{x^2} \sec^2 t \, dt}{x \sin x}$ is -

(A) 0 (B) 3
(C) 2 (D) 1

## Sum of Series using Definite Integration

**Q29.** $\lim_{n\to\infty} \frac{1^P + 2^P + 3^P + .... + n^P}{n^{P+1}}$ equals -

(A) 1 (B) $\frac{1}{P+1}$

(C) $\frac{1}{P+2}$ (D) $P^2$

**Q30.** $\lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{n} e^{r/n}$ is-

(A) e (B) e – 1
(C) 1 – e (D) e + 1

## Reduction Formula

**Q31.** If $I_n = \int_0^{\infty} e^{-x} x^{n-1}\,dx$ then $\int_0^{\infty} e^{-\lambda x} x^{n-1}\,dx =$

(A) $\lambda I_n$ (B) $\frac{I_n}{\lambda}$

(C) $\frac{I_n}{\lambda^n}$ (D) $\lambda^n I_n$

**Q32.** If $u_{10} = \int_0^{\pi/2} x^{10} \sin x\,dx$, then the value of $u_{10} + 90u_8$ is-

(A) $9\left(\frac{\pi}{2}\right)^8$ (B) $\left(\frac{\pi}{2}\right)^9$

(C) $10\left(\frac{\pi}{2}\right)^9$ (D) $9\left(\frac{\pi}{2}\right)^9$

# *Exercise - 1A* *Concept Builder*

## ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

**Q1.** If $\frac{2x}{\pi} < \sin x < x$ for $0 < x < \frac{\pi}{2}$, then the value of the integral $\int_0^{\pi/2} \frac{\sin x}{x}\,dx$ is

(A) greater than 1 (B) less than 1

(C) greater than $\frac{\pi}{2}$ (D) less than $\frac{\pi}{2}$

**Q2.** If $f(x) = \int_0^x (\cos^4 t + \sin^4 t)\,dt$ then $f(x + \pi)$ will be equal to

(A) $f(x) + f(\pi)$ (B) $f(x) + 2f(\pi)$

(C) $f(x) + f\left(\frac{\pi}{2}\right)$ (D) $f(x) + 2f\left(\frac{\pi}{2}\right)$

**Q3.** The value of $\int_{-\pi}^{\pi} \sin mx \cos n x\,dx\,(m, n \in N)$ is equal to

(A) 0

(B) $\int_{-\pi/8}^{\pi/8} x^8 \sin^9 x\,dx$

(C) $\int_{-\pi/2}^{\pi/2} \sin x\, f(\cos x)\,dx$

(D) $\int_{-1/2}^{1/2} e^{\cos x}\,dx$

**Q4.** If x satisfies the equation

$x^2\left(\int_0^{\pi/2} (2\sin t + 3\cos t)\,dt\right) - x\left(\int_{-3}^{3} \frac{t^2 \sin 2t}{t^2+1}\right) - 2 = 0$, then the value of x is-

(A) $-1$ (B) 1

(C) $-\sqrt{2/5}$ (D) $\sqrt{2/5}$

**Q5.** If $I = \int_0^{\pi/2} \log \sin x\,dx$ then which of the following integrals are also equal to $I$

(A) $\int_0^{\pi/2} \log \sin 2x\,dx$

(B) $\int_0^{\pi} \log \sqrt{\sin x}\,dx$

(C) $\int_0^{\pi} \log (1 + \cos x)\,dx$

(D) $\int_0^{\pi/2} \log \cos x\,dx$

**Q6.** If $I = \int_0^{2\pi} \sin^2 x\,dx$, then

(A) $I = 2\int_0^{\pi} \sin^2 x\,dx$

(B) $I = 4\int_0^{\pi/2} \sin^2 x\,dx$

(C) $I = \int_0^{2\pi} \cos^2 x\,dx$

(D) $I = 8\int_0^{\pi/4} \sin^2 x\,dx$

**Q7.** Which of the following is/are true-

(A) $\int_0^{n\pi} f(\cos^2 x)dx = n\int_0^{\pi} f(\cos^2 x)\,dx \quad n \in N$

(B) $\int_0^{b-c} f(x+c)\,dx = \int_b^{c} f(x)\,dx$

(C) $\int_a^{\pi-a} x f(\sin x)\,dx = \frac{\pi}{2}\int_a^{\pi-a} f(\sin x)\,dx$

(D) $\int_0^{\pi} x f(\sin^3 x + \cos^2 x)\,dx$

$= 2\pi \int_0^{\pi/2} f(\sin^3 x + \cos^2 x)\,dx$

**Q8.** The points of extremum of $\int_0^{x^2} \frac{t^2 - 5t + 4}{2 + e^t}\,dt$ are-

(A) $x = -2$ (B) $x = 1$

(C) $x = 0$ (D) $x = -1$

**Q9.** $\lim_{x \to 0} = \frac{xe^{x^2}}{\int_0^x e^{t^2}\,dt}$

(A) 0 (B) 1

(C) –1 (D) does not exist

## PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

**Paragraph # 1 (Ques. 10 to 12)**

A polynomial $P(x) = ax^3 + bx^2 + cx + d$ vanishes at $x = 1$ and has relative maximum/minimum at $x = 1$ and $x = -2$. If $\int_{-1}^{1} P(x)\,dx = 16$, then :

**Q10.** The nature of roots of $P(x) = 0$ is -

(A) One real and two complex

(B) All real and distinct

(C) All real but two of those are equal

(D) All real with two of those are negative

**Q11.** The set of values of a such that the equation $P(x) - a = 0$ has exactly three distinct real roots is -

(A) (0, 27) (B) (–2, 3)

(C) (0, 9) (D) (0, 8)

**Q12.** $\int_{-2}^{2} P(x)\,dx$ is -

(A) 11 (B) 22

(C) 33 (D) 44

**Paragraph # 2 (Ques. 13 to 15)**

Let $f(x) = ax^2 + bx + c$ is quadratic polynomial, where a, b, c $\in$ R and $a > 0$, $f(x) = 0$ has two real and different positive roots $\alpha$ and $\beta$ ($\alpha < \beta$).

Now $\int_0^{\alpha} f(x)\,dx = A$ and $\int_{\alpha}^{\beta} f(x)\,dx = B$.

**Q13.** Which statement is correct-

(A) $A > 0, B > 0$ (B) $A < 0, B < 0$

(C) $A > 0, B < 0$ (D) $A < 0, B > 0$

**Q14.** Find $\int_{-\beta}^{0} f(|x|)\,dx$

(A) A + B (B) 2A – 2B

(C) A – B (D) none of these

**Q15.** Find $\int_{-\beta}^{\beta} (f(|x|)+|f(|x|)|)\,dx$

(A) 4A (B) 2B

(C) 2(A + B) (D) none of these

**Paragraph # 3 (Ques. 16 to 18)**

Let there are two function

$f(x) = \min(|x-1|, |x|, |x+1|)$ and

$g(x) = \max(\sin^{-1}(\sin x), \cos^{-1}(\cos x))$

**Q16.** Find the no. of non differential points of f(x) in $x \in (-1, 1)$

(A) 2 (B) 3

(C) 1 (D) none of these

**Q17.** Which statement is correct -

(A) $\int_0^1 f(x)\,dx = \int_0^2 g(x)\,dx$

(B) $\int_{-1/2}^{1/2} f(x)\,dx = \int_{-1/2}^{1/2} g(x)\,dx$

(C) $\int_0^1 f(x)\,dx > \int_0^{\pi/2} g(x)\,dx$

(D) none of these

**Q18.** Find $\int_0^{2\pi} g(x)\,dx$

(A) $\pi^2$ (B) $2\pi^2$

(C) $\frac{\pi^2}{2}$ (D) none of these

## MATCH THE COLUMN

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

**Q19.** Let $I_1 = \int_{-1}^{1} f(x)\,dx$, $I_2 = \int_{-1}^{1} g(x)\,dx$, & $I_3 = \int_{-1}^{1} h(x)\,dx$, be three integrals, match the following columns accordingly.

| **Column I** | **Column II** |
|---|---|
| (A) $f(x) = e^x, g(x) = e^{-x}, h(x) = e^{\|x\|}$ | (P) $I_1 < I_2$ |
| (B) $f(x) = \|x\|, g(x) = x^2, h(x) = x^3$ | (Q) $I_2 < I_3$ |
| (C) $f(x) = x, g(x) = \|x\|^3, h(x) = x^2$ | (R) $I_1 < I_3$ |
| (D) $f(x) = e^{-\|x\|}, g(x) = e^x, h(x) = e^{-x}$ | (S) $I_1 > I_2$ |

**Q20.** Match the integrals in Column-I with the values in Column-II.

| **Column-I** | **Column-II** |
|---|---|
| (A) $\int_{-1}^{1} \frac{dx}{1+x^2}$ | (P) $\frac{1}{2}\log\left(\frac{2}{3}\right)$ |
| (B) $\int_{0}^{1} \frac{dx}{\sqrt{1-x^2}}$ | (Q) $2\log\left(\frac{2}{3}\right)$ |
| (C) $\int_{2}^{3} \frac{dx}{1-x^2}$ | (R) $\frac{\pi}{3}$ |
| (D) $\int_{1}^{2} \frac{dx}{x\sqrt{x^2-1}}$ | (S) $\frac{\pi}{2}$ |

**Q21.** If f is an odd function then -

| **Column-I** | **Column-II** |
|---|---|
| (A) $\int_{-a}^{a} (f(x))^2\,dx$ | (P) 0 |
| (B) $\int_{-a}^{a} x(f(x))^2\,dx$ | (Q) $\int_{0}^{a} (f(x))^4 + (f(a-x))^4\,dx$ |
| (C) $\int_{-a}^{a} (x^2+x^3)(f(x))^2\,dx$ | (R) $2\int_{0}^{a} (f(x))^2\,dx$ |
| (D) $\int_{-a}^{a} (f(x))^4\,dx$ | (S) $2\int_{0}^{a} x^2(f(x))^2 dx$ |

**Q22.**

| **Column-I** | **Column-II** |
|---|---|
| (A) $\int_{0}^{\frac{\pi}{2}} e^{\sin^2 x}.\sin 2x\,dx$ | (P) $\frac{\pi}{4}$ |
| (B) $\int_{-1}^{1} x\,\lvert x\rvert\,dx$ | (Q) e – 1 |
| (C) $\int_{0}^{\frac{\pi}{2}} \frac{(\sin x)^{\frac{5}{2}}}{(\sin x)^{\frac{5}{2}}+(\cos x)^{\frac{5}{2}}}dx$ | (R) $\frac{\pi}{32}$ |
| (D) $\int_{0}^{\frac{\pi}{2}} \sin^4 x\cos^2 x\;dx =$ | (S) $\frac{\pi}{2}$ |
| | (T) 0 |

## NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

**Q23.** Evaluate: $\int_0^{\infty} \frac{\ln(1+x+\sqrt{x(1+x)})}{(x+1)\cdot\sqrt{x(1+x)}}dx$

**Q24.** If $\int_0^{10} f(x)\,dx = 5$, find the value of

$$\sum_{k=1}^{10}\int_0^1 f(k-1+x)\,dx.$$

**Q25.** $\int_0^{\pi/2} \left(\sqrt{\sin x}+\sqrt{\cos x}\right)^{-4} dx$

**Q26.** If the value of the definite integral

$$I = \int_0^2 (3x^2-3x+1)\cos(x^3-3x^2+4x-2)\,dx$$

can be expressed in the form as p(sin q), where p, q ∈ N, then find (p + q).

**Q27.** $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x}dx$

**Q28.** Evaluate $\lim\limits_{x\to 0} \frac{\int_0^x \tan x\,dx}{x\tan x}$.

**Q29.** If x = a is the point of maxima of

$$f(x) = \int_0^{x^2} \frac{t^2-t}{e^t+1}dt.$$ Find 'a'.

**Q30.** Evaluate

$$\lim_{n\to\infty} n\left[\frac{1}{(n+1)^2}+\frac{1}{(n+2)^2}+.....+\frac{1}{(2n)^2}\right].$$

**Q31.** Let $I_n = \int_0^{\frac{\pi}{4}} \tan^n x\,dx$ $(n=0, 1, 2, 3, 4,.....)$

and

$$S_n = \sum_{n=0}^{n}\left(I_nI_{n+1}+I_nI_{n+3}+I_{n+1}I_{n+2}+I_{n+2}I_{n+3}\right).$$

Find the value of $\lim\limits_{n\to\infty} 100(S_n)$.

**Q32.** For a positive integer n,

let $I_n = \int_{-\pi}^{\pi} \left(\frac{\pi}{2}-|x|\right)\cos nx\,dx$.

Find the value of $9(I_1+I_2+I_3+I_4)$.

**Q33.** Let $u = \int_0^{\pi/4} \left(\frac{\cos x}{\sin x+\cos x}\right)^2 dx$

and $v = \int_0^{\pi/4} \left(\frac{\sin x+\cos x}{\cos x}\right)^2 dx$.

Find the value of $\frac{v}{u}$.

## SUBJECTIVE TYPE

**Q34.** $\int_1^2 x^2\,dx$

**Q35.** $\int_0^2 \frac{dx}{4+x^2}$

**Q36.** $\int_0^1 xe^{x^2}dx$

**Q37.** $\int_0^{\pi/2} \sin^2 x\,dx$

**Q38.** $\int_1^2 \frac{1-x}{1+x}dx$ equals-

**Q39.** $\int_0^{\infty} \frac{\sin(\tan^{-1} x)}{1+x^2}dx$

**Q40.** If $f(x) = \begin{cases} x^2, & 0 < x < 2 \\ 3x-4, & 2 \le x < 3 \end{cases}$ then evaluate $\int_0^3 f(x)dx$.

**Q41.** Evaluate $\int_0^2 |1-x|\,dx$.

**Q42.** Evaluate $\int_0^{\pi/2} \sin^6 x\,dx$.

**Q43.** Evaluate $\int_0^{\pi/2} \cos^7 x\,dx$.

**Q44.** Evaluate $\int_0^{\pi/2} \sin^4 x \cos^5 x\,dx$.

**Q45.** $\int_0^{\pi/2} \frac{dx}{1+\cot x}$

**Q46.** Evaluate : $\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}}$

**Q47.** $\int_0^1 \log\left(\frac{1}{x}-1\right)dx$

**Q48.** $\int_0^{\pi/2} \log \tan x\,dx$

**Q49.** Find $\frac{d}{dt}\int_{t^2}^{t^3} \frac{1}{\log x}dx$

**Q50.** Evaluate $\lim_{n\to\infty}\left[\frac{1}{n+1}+\frac{1}{n+2}+...+\frac{1}{2n}\right]$

## *Exercise - 2* — *Brain Booster*

### SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer.* ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

**Q1.** Let a, b, c be non-zero real numbers such that ; $\int_0^1 (1+\cos^8 x)(ax^2+bx+c)dx = \int_0^2 (1+\cos^8 x)(ax^2+bx+c)dx$

then the quadratic equation $ax^2 + bx + c = 0$ has -

(A) no root in (0,2)

(B) atleast one root in (0,2)

(C) a double root in (0,2)

(D) none

**Q2.** The value of $\int_{-2}^{3} |1-x^2|dx$ is-

(A) 28/3 (B) 14/3

(C) 7/3 (D) 1/3

**Q3.** $\int_0^1 \sqrt{\frac{1-x}{1+x}}\,dx =$

(A) $\frac{\pi}{2}+1$ (B) $\frac{\pi}{2}-1$

(C) $\pi$ (D) 1

**Q4.** $\int_{-2}^{0} (x^3+3x^2+3x+3+(x+1)\cos(x+1))\,dx$

is equal to -

(A) $-4$ (B) 0

(C) 4 (D) 6

**Q5.** The integral $\int_{-\frac{1}{2}}^{\frac{1}{2}} \left([x]+\ell n\left(\frac{1+x}{1-x}\right)\right)dx$ equals,

where [ ] denotes greatest integer function -

(A) $-\frac{1}{2}$ (B) 0

(C) 1 (D) $2\ell n\left(\frac{1}{2}\right)$

**Q6.** If $I_1=\int_0^1 2^{x^2}\,dx$, $I_2=\int_0^1 2^{x^3}\,dx$, $I_3=\int_1^2 2^{x^2}\,dx$

and $I_4=\int_1^2 2^{x^3}\,dx$ then-

(A) $I_2>I_1$ (B) $I_1>I_2$

(C) $I_3=I_4$ (D) $I_3>I_4$

**Q7.** The value of $\int_1^a [x]f'(x)\,dx$, $a>1$, where [x] denotes the greatest integer not exceeding x is-

(A) [a] f(a) – {f(1) + f(2) + ... + f([a])}

(B) [a] f([a]) – {f(1) + f(2) + ... + f(a)}

(C) a f([a]) – {f(1) + f(2) + ... + f(a)}

(D) a f(a) – {f(1) + f(2) + ... + f([a])}

**Q8.** Let $F(x)=f(x)+f\left(\frac{1}{x}\right)$,

where $f(x)=\int_1^x \frac{\log t}{1+t}\,dt$.

Then F(e) equals-

(A) $\frac{1}{2}$ (B) 0 (C) 1 (D) 2

**Q9.** $\int_3^4 \sqrt{(4-x)(x-3)}\,dx =$

(A) $\frac{1}{3}$ (B) 1 (C) 0 (D) $\frac{\pi}{8}$

**Q10.** Let $\frac{d}{dx}f(x)=\left(\frac{e^{\sin x}}{x}\right)$, $x>0$.

If $\int_1^4 \frac{3}{x}e^{\sin x^3}\,dx = f(k) - f(1)$, then one of the possible values of k, is-

(A) 64 (B) 15

(C) 16 (D) 63

**Q11.** If $f(y)=e^y$, $g(y)=y$; $y>0$ and

$F(t)=\int_0^t f(t-y)\,g(y)\,dy$, then-

(A) $F(t)=te^{-t}$

(B) $F(t)=1-e^{-1}(1+t)$

(C) $F(t)=e^t-(1+t)$

(D) $F(t)=te^t$

**Q12.** Evaluate : $\int_{-\pi/3}^{\pi/3} \frac{\pi+4x^3}{2-\cos\left(|x|+\frac{\pi}{3}\right)}\,dx$

(A) $\frac{2\pi}{\sqrt{3}}\tan^{-1}(1/2)$ (B) $\frac{8\pi}{\sqrt{3}}\tan^{-1}(1/2)$

(C) $\frac{4\pi}{\sqrt{3}}\tan^{-1}(1/2)$ (D) $\frac{16\pi}{\sqrt{3}}\tan^{-1}(1/2)$

**Q13.** The value of $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x} dx$, $a > 0$ is-

(A) $a\pi$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{a}$ (D) $2\pi$

**Q14.** The value of the integral $I = \int_0^1 x(1-x)^n dx$ is-

(A) $\frac{1}{n+1} + \frac{1}{n+2}$ (B) $\frac{1}{n+1}$

(C) $\frac{1}{n+2}$ (D) $\frac{1}{n+1} - \frac{1}{n+2}$

**Q15.** $\int_2^3 \frac{(x+2)^2}{2x^2-10x+53} dx$ is equal to -

(A) 2 (B) 1 (C) 1/2 (D) 5/2

**Q16.** If $f(x) = \frac{e^x}{1+e^x}$, $I_1 = \int_{f(-a)}^{f(a)} xg\{x(1-x)\} dx$ and $I_2 = \int_{f(-a)}^{f(a)} g\{x(1-x)\} dx$, then the value of $\frac{I_2}{I_1}$ is-

(A) 2 (B) –3 (C) –1 (D) 1

**Q17.** Evaluate

$$\int_0^{\pi} e^{|\cos x|} \left(2\sin\left(\frac{1}{2}\cos x\right) + 3\cos\left(\frac{1}{2}\cos x\right)\right) \sin x \, dx$$

(A) $\frac{24}{5}\left[e\cos(1/2) + \frac{1}{2}e\sin\left(\frac{1}{2}\right) - 1\right]$

(B) $\frac{24}{5}\left[e\cos(1/2) + \frac{1}{2}e\sin\left(\frac{1}{4}\right) - 1\right]$

(C) $\frac{24}{5}\left[e\cos(1/2) + \frac{1}{4}e\sin\left(\frac{1}{2}\right) - 1\right]$

(D) $\frac{24}{5}\left[e\cos(1/4) + \frac{1}{2}e\sin\left(\frac{1}{2}\right) - 1\right]$

**Q18.** Let $T > 0$ be a fixed real number. Suppose f is a continuous function such that for all $x \in R$ $f(x+T) = f(x)$. If $I = \int_0^T f(x) dx$ then the value of $\int_3^{3+3T} f(2x) dx$ is -

(A) $\frac{3}{2}I$ (B) 2I

(C) 3I (D) 6I

**Q19.** For $n \in N$, the value of the definite integral $\int_0^{n\pi+V} \sqrt{\frac{1+\cos 2x}{2}} dx$ where $\frac{\pi}{2} < V < \pi$ is -

(A) 2n + 1 – cosV (B) 2n – sinV

(C) 2n + 2 – sinV (D) 2n + 1 – sinV

**Q20.** If $x > 0$, $\int_0^{100[x]} (2x - [2x])^2 dx$ is equal to, (where [ ] denotes greatest integer function)-

(A) $\frac{100[x]}{3}$ (B) $\frac{100\{x\}}{3}$

(C) $\frac{100x}{3}$ (D) 200

**Q21.** If $\int_0^{t^2} x f(x) dx = \frac{2}{5}t^5$, $t > 0$, then $f\left(\frac{4}{25}\right) =$

(A) $\frac{2}{5}$ (B) $\frac{5}{2}$ (C) $-\frac{2}{5}$ (D) 1

**Q22.** $\lim_{x \to \frac{\pi}{4}} \frac{\int_2^{\sec^2 x} f(t)dt}{x^2 - \frac{\pi^2}{16}}$ equals -

(A) $\frac{8}{\pi}f(2)$ (B) $\frac{2}{\pi}f(2)$

(C) $\frac{2}{\pi}f\left(\frac{1}{2}\right)$ (D) 4f(2)

**Q23.** If $\int_{\sin x}^{1} t^2 (f(t))\,dt = (1-\sin x)$ then $f\left(\frac{1}{\sqrt{3}}\right)$ is -

(A) 1/3 (B) $1/\sqrt{3}$

(C) 3 (D) $\sqrt{3}$

**Q24.** Let $f(x) = \int_{1}^{x} \sqrt{2-t^2}\,dt$. Then the real roots of the equation $x^2 - f'(x) = 0$ are -

(A) $\pm 1$ (B) $\pm\frac{1}{\sqrt{2}}$

(C) $\pm\frac{1}{2}$ (D) 0 and 1

**Q25.** If $\int_{0}^{f(x)} t^2 dt = x\cos\pi x$, then $f'(9)$

(A) is equal to $-\frac{1}{9}$ (B) is equal to $-\frac{1}{3}$

(C) is equal to $\frac{1}{3}$ (D) is non existent

**Q26.** If $g(x) = \int_{0}^{x} \cos^4 t\,dt$, then $g(x+\pi)$ equals -

(A) $g(x) + g(\pi)$ (B) $g(x) - g(\pi)$

(C) $g(x)g(\pi)$ (D) $\frac{g(x)}{g(\pi)}$

**Q27.** $\lim_{n\to\infty}\left[\frac{1}{n}+\frac{1}{\sqrt{n^2+n}}+\frac{1}{\sqrt{n^2+2n}}+\ldots+\frac{1}{\sqrt{n^2+(n-1)n}}\right]$ is equal to-

(A) $2+2\sqrt{2}$ (B) $2\sqrt{2}-2$

(C) $2\sqrt{2}$ (D) 2

**Q28.** $\lim_{n\to\infty}\left[\frac{n+1}{n^2+1^2}+\frac{n+2}{n^2+2^2}+\ldots\ldots+\frac{1}{n}\right]$ equals-

(A) $\frac{\pi}{4}+\frac{1}{2}\log 2$ (B) $\frac{\pi}{4}-\frac{1}{2}\log 2$

(C) $\frac{\pi}{4}-2\log\frac{1}{2}$ (D) none of these

**Q29.** $\lim_{n\to\infty}\left[\frac{1}{n^2}\sec^2\frac{1}{n^2}+\frac{2}{n^2}\sec^2\frac{4}{n^2}+\ldots+\frac{1}{n}\sec^2 1\right]$ equals-

(A) $\frac{1}{2}\sec 1$ (B) $\frac{1}{2}\operatorname{cosec} 1$

(C) $\tan 1$ (D) $\frac{1}{2}\tan 1$

**Q30.** If $I_n = \int_{0}^{\pi/4} \tan^n x\,dx$, $n \in N$, then $I_{n+2} + I_n$ equals-

(A) $\frac{1}{n}$ (B) $\frac{1}{n-1}$

(C) $\frac{1}{n+1}$ (D) $\frac{1}{n+2}$

**Q31.** Let $f(x) = \frac{x}{(1+x^n)^{1/n}}$ for $n \geq 2$ and

$g(x) = \underbrace{(f\circ f\circ\ldots\circ f)}_{\text{f occurs n times}}(x)$.

Then $\int x^{n-2} g(x)dx$ equals.

(A) $\frac{1}{n(n-1)}(1+nx^n)^{1-\frac{1}{n}}+K$

(B) $\frac{1}{n-1}(1+nx^n)^{1-\frac{1}{n}}+K$

(C) $\frac{1}{n(n+1)}(1+nx^n)^{1+\frac{1}{n}}+K$

(D) $\frac{1}{n+1}(1+nx^n)^{1+\frac{1}{n}}+K$

## ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

**Q1.** The value of the integral

$$\int_0^{\pi/4} \frac{dx}{(a^2\cos^2 x + b^2\sin^2 x)} \text{ is}$$

(A) $\frac{1}{ab}\tan^{-1}\left(\frac{b}{a}\right)(a>0,\ b>0)$

(B) $\frac{1}{ab}\tan^{-1}\left(\frac{b}{a}\right)(a<0, b<0)$

(C) $\frac{\pi}{4}(a=1, b=1)$

(D) None of these

**Q2.** The integral $\int_0^{\pi} xf(\sin x)dx$ as equal to

(A) $\frac{\pi}{2}\int_0^{\pi} f(\sin x)dx$ (B) $\frac{\pi}{4}\int_0^{\pi} f(\sin x)dx$

(C) $\pi\int_0^{\pi/2} f(\sin x)dx$ (D) $\pi\int_0^{\pi/2} f(\cos x)dx$

**Q3.** If x satisfies the equation

$$x^2\left(\int_0^1 \frac{dt}{t^2+2t\cos\alpha+1}\right) - x\left(\int_{-3}^{3}\frac{t^2\sin 2t}{t^2+1}dt\right) - 2 = 0,$$

$(0<\alpha<\pi)$, then the value of x is

(A) $2\sqrt{\left(\frac{\sin\alpha}{\alpha}\right)}$ (B) $-2\sqrt{\left(\frac{\sin\alpha}{\alpha}\right)}$

(C) $4\sqrt{\left(\frac{\sin\alpha}{\alpha}\right)}$ (D) None of these

**Q4.** The value of $\int_a^{a+\pi/2} (\sin^4 x + \cos^4 x)dx$ is

(A) independent of a

(B) $a\left(\frac{\pi}{2}\right)^2$

(C) $3\pi/8$

(D) $\frac{3}{8}\pi a^2$

**Q5.** Let f(x) be a continuous function for all x in its domain, which is not identically zero such that $\{f(x)\}^2 = \int_0^x f(t)\frac{2\sec^2 t}{4+\tan t}dt$.

(A) $f\left(\frac{\pi}{4}\right) = \ell n\frac{5}{4}$

(B) $f\left(\frac{\pi}{2}\right) = \ell n\, 4$

(C) $f\left(\frac{3\pi}{4}\right) = \ell n\frac{3}{4}$

(D) f(x) = 0 has infinite roots

**Q6.** Let $S_n = \sum_{k=1}^{n}\frac{n}{n^2+kn+k^2}$ and

$T_n = \sum_{k=0}^{n-1}\frac{n}{n^2+kn+k^2}$ for n = 1, 2, 3, .....

Then,

(A) $S_n < \frac{\pi}{3\sqrt{3}}$ (B) $S_n > \frac{\pi}{3\sqrt{3}}$

(C) $T_n < \frac{\pi}{3\sqrt{3}}$ (D) $T_n > \frac{\pi}{3\sqrt{3}}$

**Q7.** If $I_n = \int_0^{\pi/4} \tan^n x \, dx$ then

(A) $I_7 + I_5 = \frac{1}{6}$

(B) $I_{10} + I_8 = \frac{1}{9}$

(C) $I_8 - I_{12} = \frac{2}{99}$

(D) $I_{12} + 2I_{10} + I_8 = \frac{20}{99}$

## PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

### Paragraph # 1 (Ques. 8 to 11)

If function f(x) is continuous in the interval (a, b) and having same definition between a and b, then we can find $\int_a^b f(x)\,dx$ and if f(x) is discontinuous and not having same definition between a and b, then we must break the interval such that f(x) becomes continuous and having same defintion in the breaking intervals.

Now, if f(x) is discontinuous at x = c(a < c < b),

then $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$ and also if f(x) is discontinuous at x = a in (0, 2a), then we can write $\int_0^{2a} f(x)\,dx = \int_0^a \{f(a-x) + f(a+x)\}\,dx$.

**Q8.** $\int_{-1}^{1} [3x-1]\,dx$ (where [.] denotes greatest integer function) is equal to-

(A) –5 (B) –3 (C) –1 (D) None

**Q9.** $\int_{\pi/2}^{3\pi/2} [2\sin x]\,dx$ (where [.] denotes the greatest integer function) is equal to-

(A) $-\pi/2$ (B) $-\pi$

(C) 0 (D) $\pi/2$

**Q10.** $\int_0^{10} \left[\frac{x^2+2}{x^2+1}\right] dx$ (where [.] denotes the greatest integer function) is equal to-

(A) 0 (B) 2

(C) 5 (D) 10

**Q11.** $\int_0^1 \sin([x]+[2x])\,dx$ (where [.] denotes the greatest integer function) is equal to-

(A) sin 1 (B) $\sin\left(\frac{3}{2}\right)$

(C) $\frac{\sin 1}{2}$ (D) $\frac{\sin 2}{3}$

### Paragraph # 2 (Ques. 12 to 14)

If f(x) and g(x) be two functions, such that f(a) = g(a) = 0 and f and g are both differentiable at everywhere in some neighbourhood of point a except possibly 'a'.

Then $\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$, provided f '(a) and g '(a) are not both zero.

**Q12.** The value of $\lim_{x\to 0} \frac{\int_0^{x^2} \sin\sqrt{t}\,dt}{x^3}$ is (x > 0)

(A) 0 (B) 2/9 (C) 1/3 (D) 2/3

**Q13.** The value of $\lim_{x\to\infty} \frac{\left\{\int_0^x e^{t^2} dt\right\}^2}{\int_0^x e^{2t^2} dt}$ is

(A) 1/3 (B) 2/3

(C) 1 (D) None of these

**Q14.** The value of $\lim_{t\to 0} \frac{\int_0^t x\,dx}{t\sin t}$ is

(A) 1/4 (B) 1/3

(C) 1/2 (D) 0

**Paragraph # 3 (Ques. 15 to 17)**

Let f(x) be a continuous function defined on the closed interval [a, b], then

$$\lim_{n\to\infty}\sum_{r=0}^{n-1}\frac{1}{n}f\left(\frac{r}{n}\right)=\int_0^1 f(x)\,dx\,.$$

**Q15.** The value of

$$\lim_{n\to\infty}\frac{1}{n}\left\{\frac{1}{n+1}+\frac{2}{n+2}+.....+\frac{3n}{4n}\right\}\text{ is}$$

(A) $5-2\,\ell n\,2$ (B) $4-2\,\ell n\,2$

(C) $3-2\,\ell n\,2$ (D) $2-2\,\ell n\,2$

**Q16.** The value of

$$\lim_{n\to\infty}\left\{\frac{1}{n}+\frac{n^2}{(n+1)^3}+\frac{n^2}{(n+2)^3}+.....+\frac{1}{8n}\right\}\text{ is}$$

(A) 5/4 (B) 3/4

(C) 5/8 (D) 3/8

**Q17.** The value of $\lim_{n\to\infty}\left\{\prod_{r=1}^{n}\left(1+\frac{r}{n}\right)\right\}^{1/n}$ is

(A) 3/e (B) 4/e

(C) 1/e (D) 2/e

## MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

**Q18.**

| **Column I** | **Column II** |
|---|---|
| (A) $f(x)=\max\left(\sin x,\cos x,\frac{1}{\sqrt{2}}\right)$ | (P) f(x) is continuous at x = 0 |
| (B) $f(x)=\min(x, x^3)$ | (Q) f(x) is differentiable at x = 0 |
| (C) $f(x)=\max((x-1)^2, 2x-x^2)$ | (R) $\int_0^{\pi} f(x)\,dx=2$ |
| (D) $f(x)=\min(\lvert x\rvert, \sin x)$ | (S) $\int_{-1}^{1} f(x)\,dx=-\frac{1}{2}$ |

**Q19.** [x] and {x} are greatest and fractional part of real x.

| **Column I** | **Column II** |
|---|---|
| (A) $\int_1^2 \frac{dx}{x^2-2x[x]+1+[x]}=$ | (P) $\frac{\pi}{4}$ |
| (B) $\int_{-(\pi/4)}^{0}\{\sin x+\cos x\}\,dx=$ | (Q) $\int_1^2 \frac{dx}{x^2-2x+2}$ |
| (C) If $\sin^{-1}\frac{4x}{\sqrt{2}\pi}-\cos^{-1}\frac{4x}{\sqrt{2}\pi}=0$ then x is equal to- | (R) $\int_{-(\pi/4)}^{0}(\sin x+\cos x)\,dx$ |
| (D) $\int_0^{\pi/4}\left\{\sqrt{2}\sin\left(x+\frac{\pi}{4}\right)\right\}dx+\int_0^{1/\sqrt{2}}\frac{dx}{\sqrt{1-x^2}}=$ | (S) $\int_0^{\pi/4}\left(\sqrt{2}\sin\left(x+\frac{\pi}{4}\right)-1\right)dx$ |

**Q20.** Match the column -

| Column I | Column II |
|---|---|
| (A) f(x) is a continuous function such that $f(x) \geq 0$; $x \in [2, 10]$ and, $\int_4^8 f(x)\,dx = 0$ then f(6) is | (P) 2 |
| (B) $\lim_{n\to\infty}$ is $\sum_{r=0}^{n-1} \frac{1}{\sqrt{4n^2 - r^2}}$ | (Q) $\pi/2$ |
| (C) f(x) is an odd function, then $\int_{-\pi/6}^{\pi/6} \frac{f(\sin x)}{f(\cos x) + f(\sin^2 x)}$ | (R) $\pi/6$ |
| (D) If $n \in I$, then $\int_{-\pi/6}^{\pi/6} \frac{\sin^{2n-1} x}{\cos^n x}$ is | (S) 0 |

## NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

**Q21.** Suppose that f, f ' and f " are continuous on $[0, \ln 2]$ and that $f(0) = 0$, $f'(0) = 3$, $f(\ln 2) = 6$, $f'(\ln 2) = 4$ and $\int_0^{\ln 2} e^{-2x} \cdot f(x)\,dx = 3$.

Find the value of $\int_0^{\ln 2} e^{-2x} \cdot f''(x)\,dx$.

**Q22.** Suppose that the function f, g, f' and g' are continuous over [0, 1], $g(x) \neq 0$ for` $x \in [0, 1]$, $f(0) = 0$, $g(0) = \pi$, $f(1) = \frac{2009}{2}$ and $g(1) = 1$.

Find the value of the definite integral,

$$\int_0^1 \frac{f(x)\cdot g'(x)\{g^2(x)-1\} + f'(x)\cdot g(x)\{g^2(x)+1\}}{g^2(x)}\,dx.$$

## SUBJECTIVE TYPE

**Q23.** $\int_0^5 |x^2 - 5x + 6|\,dx$

**Q24.** $\int_{-1}^{2} \left[x + \frac{1}{2}\right] dx$

**Q25.** $\int_0^{\pi} \text{sgn}(1 - 2\sin x)\,dx$

**Q26.** $\int_0^1 \frac{\sin^{-1}\sqrt{x}}{\sqrt{x\,(1-x)}}\,dx$

**Q27.** Given $f'(x) = \frac{\cos x}{x}$, $f\left(\frac{\pi}{2}\right) = a$, $f\left(\frac{3\pi}{2}\right) = b$.

Find the value of the definite integral $\int_{\pi/2}^{3\pi/2} f(x)\,dx$.

**Q28.** $\int_2^e \left(\frac{1}{\ell n x} - \frac{1}{\ell n^2 x}\right) dx$

**Q29.** $\int_2^3 \frac{dx}{\sqrt{(x-1)(5-x)}}$

**Q30.** $\int_1^e \frac{dx}{\ln(x^x e^x)}$

**Q31.** $\int_0^1 \frac{dx}{x^2+2x\cos\alpha+1}$ where $-\pi < \alpha < \pi$

**Q32.** $\int_0^1 x\, f''(x)\, dx$, where $f(x) = \cos(\tan^{-1}x)$

**Q33.** $\int_0^{\pi/2} \frac{dx}{1+\cos\theta \cdot \cos x}$ $\theta \in (0, \pi)$

**Q34.** $\int_0^3 \sqrt{\frac{x}{3-x}}\, dx$

**Q35.** $\int_0^2 [x^2 - x + 1]dx$, $[\cdot]$ is the greatest integer function.

**Q36.** $\int_0^3 \frac{dx}{\sqrt{1+x}+\sqrt{1+5x}}$

**Q37.** Let $A = \int_{3/4}^{4/3} \frac{2x^2+x+1}{x^3+x^2+x+1}dx$ then find the value of $e^A$.

**Q38.** $\int_0^1 \frac{2-x^2}{(1+x)\sqrt{1-x^2}}\, dx$

**Q39.** If 'f' is an even function, then prove that

$\int_0^{\pi/2} f(\cos 2x)\cos x\, dx = \sqrt{2}$

$\int_0^{\pi/4} f(\sin 2x)\cos x\, dx$ .

**Q40.** $\int_{1/3}^{3} \frac{\sin^{-1}\frac{x}{\sqrt{1+x^2}}}{x}dx$

**Q41.** $\int_1^{16} \tan^{-1}\sqrt{\sqrt{x}-1}\, dx$

**Q42.** $\int_0^{\pi} \frac{x\sin 2x \sin\left(\frac{\pi}{2}\cos x\right)}{2x-\pi}dx$

**Q43.** $\int_0^{2\pi} \frac{x\sin^{2n}x}{\sin^{2n}x+\cos^{2n}x}\, dx$

**Q44.** $\int_{\frac{\pi}{4}}^{n\pi-\frac{\pi}{4}} |\sin x + \cos x|\, dx$ $(n \in N)$

**Q45.** If $y(x) = \int_{\pi^2/16}^{x^2} \frac{\cos x \cdot \cos\sqrt{\theta}}{1+\sin^2\sqrt{\theta}}d\theta$, then find $\frac{dy}{dx}$ at $x = \pi$

**Q46.** Find the value of

$\lim_{n\to\infty}\left[\frac{n}{(n+1)^2}+\frac{n}{(n+2)^2}+....+\frac{1}{4n}\right]$

**Q47.** **(a)** If g (x) is the inverse of $f$ (x) and $f$ (x) has domain $x \in [1, 5]$, where $f(1) = 2$ and $f(5) = 10$ then find the value of

$\int_1^5 f(x)dx + \int_2^{10} g(y)dy$ .

**(b)** Suppose f is continuous, $f(0) = 0$,

$f(1) = 1$, $f'(x) > 0$ and $\int_0^1 f(x)dx = \frac{1}{3}$.

Find the value of the definite integral

$\int_0^1 f^{-1}(y)dy$ .

# Answer Key

## Ex - 1

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| B | B | D | A | D | C | B | A | C | C |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| A | B | A | C | B | A | C | A | D | A |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| A | A | C | B | B | C | B | D | B | B |
| 31 | 32 | | | | | | | | |
| C | C | | | | | | | | |

## Ex - 1A

**1.** A,D **2.** A,D **3.** A,B,C **4.** C,D **5.** A,B,D

**6.** A,B,C **7.** A,C **8.** A,B,C,D **9.** B **10.** C

**11.** A **12.** D **13.** C **14.** A **15.** A

**16.** B **17.** B **18.** A

**19.** (A)→ (Q, R) (B)→ (S); (C) → (P, Q, R); (D) → (P, R)

**20.** (A)→ (S); (B)→ (S); (C)→ (P); (D)→ (R)

**21.** (A) → (R); (B) → (P); (C) → (S); (D) → (Q)

**22.** A) → (Q); (B) → (T); (C) → (P); (D) → (R)

**23.** 2 **24.** 5 **25.** $\frac{1}{3}$ **26.** 4 **27.** $\frac{\pi}{2}$

**28.** $\frac{1}{2}$ **29.** 0 **30.** $\frac{1}{2}$ **31.** 100 **32.** 40

**33.** 4 **34.** $\frac{7}{3}$ **35.** $\frac{\pi}{8}$ **36.** $\frac{1}{2}(e-1)$ **37.** $\frac{\pi}{4}$

**38.** $2\log(3/2)-1$ **39.** 1 **40.** $\frac{37}{6}$ **41.** 1 **42.** $\frac{5}{32}\pi$

**43.** $\frac{48}{105}$ **44.** $\frac{8}{315}$ **45.** $\frac{\pi}{4}$ **46.** $\frac{\pi}{12}$ **47.** 0

**48.** 0 **49.** $\frac{t(t-1)}{\log t}$ **50.** $\log 2$

| Ex - 2 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| D | A | B | C | A | B | A | A | D | A |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| C | C | B | D | C | A | A | C | C | A |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| A | A | C | A | A | A | B | A | D | C |
| 31 | | | | | | | | | |
| A | | | | | | | | | |

## EX - 2A

**1.** $\frac{\pi}{4}$ **2.** A, C, D **3.** A, B **4.** A, C **5.** A,C,D

**6.** A, D **7.** A, B, C, D **8.** B **9.** A **10.** D

**11.** C **12.** D **13.** D **14.** C **15.** C

**16.** D **17.** B

**18.** (A) → (P,Q); (B) → (P); (C) → (P, Q); (D) → (P,Q,R)

**19.** (A) → (P, Q); (B) → (R); (C) → (P, Q); (D) → (S)

**20.** (A) → (S); (B) → (R); (C) → (S); (D) → (S)

**21.** 13 **22.** 2009 **23.** $\frac{57}{6}$ **24.** $\int_{\frac{3}{2}}^{-\frac{1}{2}} -1\,dx + \int_{-\frac{1}{2}}^{\frac{1}{2}} 0\,dx + \int_{\frac{1}{2}}^{\frac{3}{2}} dx + \int_{\frac{3}{2}}^{2} 2\,dx$

**25.** $\frac{-\pi}{3}$ **26.** $\frac{\pi^2}{4}$ **27.** $2-\frac{\pi}{2}(a-3b)$ **28.** e - $\frac{2}{\ell n 2}$ **29.** $\frac{\pi}{6}$

**30.** $l$n 2 **31.** $\frac{\alpha}{2\sin\alpha}$ if $\alpha \neq 0$ ; $\frac{1}{2}$ if $\alpha = 0$ **32.** $1-\frac{3}{2\sqrt{2}}$ **33.** $\frac{\theta}{\sin\theta}$

**34.** $\frac{3\pi}{2}$ **35.** $\frac{5-\sqrt{5}}{2}$ **36.** $1 + \frac{1}{4}\ln\left(\frac{9}{5}\right)$ **37.** $\frac{16}{9}$ **38.** $\frac{\pi}{2}$

**40.** $\frac{\pi l n 3}{2}$ **41.** $\frac{16\pi}{3} - 2\sqrt{3}$ **42.** $\frac{8}{\pi^2}$ **43.** I = $\pi^2$ **44.** $2\sqrt{2}\,n$

**45.** $2\pi$ **46.** $\frac{1}{2}$ **47.** (a) 48 ; (b) 2/3

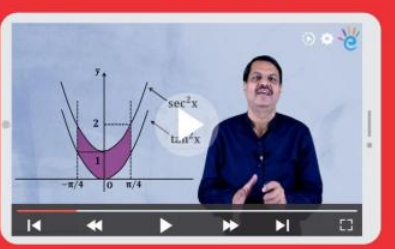
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